Electrical power and energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate electrical transfer rate and energy use.

SUCCESS CRITERIA - I can...

1. Use P=IV with watts, amps and volts.
2. Use E=Pt with seconds for joules.
3. Distinguish power from total energy.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “power” and “voltage” to explain your model or result. Sentence starters are optional.

power: rate of energy transfer

voltage: electric potential difference; energy transferred per charge

current: rate of charge flow

01 Bring back an earlier idea

Two equal 10 Ω resistors are connected in parallel.
Find equivalent resistance, then compare with a series connection of the same pair.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $P = IV$; $E = P\Delta t$; $P = I^2R = V^2/R$.
Constant power for $E = P\Delta t$; last equalities apply to an ohmic resistor.

Worked reasoning: A device at 6 V drawing 2 A receives power $VI=12$ W. Over 5 s it receives 60 J. Power is an energy-transfer rate.

Your turn: Draw an input-energy arrow and label voltage, current, power and elapsed time with distinct units.

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 Represent and interpret

Two motors use the same energy but one takes half the time.
Compare average power.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

04 Solve and justify

A 12 V device draws .75 A for 4 minutes.
Find power and energy in joules.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

Electrical power and energy

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

A student says a 10 W motor always uses more energy than a 5 W motor. Supply a time comparison that disproves this.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Choose a method

A shallow-water model uses v=√(gh). With g=10, h=40 m and travel distance 6 km, find speed and travel time.
State a model limitation.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: 1 2 3

07 Exit check

A 6 Ω resistor carries 2 A.
Find its power using two equivalent equations.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 12 V motor draws 1.5 A for 40 s and delivers 432 J of useful mechanical energy.
Find electrical input energy, average useful power and efficiency.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3